

PE-EC701C — Mobile Communication and Networks

Semester VII • 3L:0T:0P • 3 Credits

Module 4 Exam Notes: Multiple Access and Modulation Schemes

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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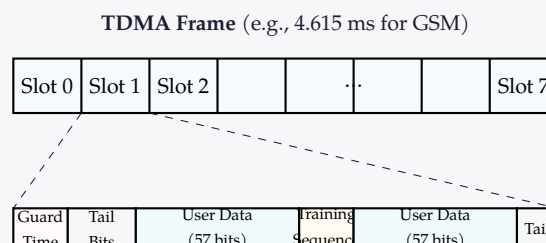
1. Multiple Access Schemes

Multiple access schemes allow many mobile users to share the same physical radio spectrum simultaneously without causing mutual interference.

1.1 FDMA, TDMA, CDMA, and SDMA

- **FDMA (Frequency Division Multiple Access):** Divides the allocated spectrum into narrow, non-overlapping frequency channels. Each user is assigned a unique channel for the duration of the call. Requires guard bands to prevent adjacent channel interference. Used in 1G systems.
- **TDMA (Time Division Multiple Access):** Divides a single carrier frequency channel into multiple non-overlapping time slots. Each user is assigned a specific slot periodically, transmitting in bursts. Requires guard times to prevent slot overlaps. Used in GSM (2G).
- **CDMA (Code Division Multiple Access):** Users share the entire allocated bandwidth simultaneously. Each user's signal is multiplied by a unique, orthogonal pseudo-noise (PN) code. The receiver uses the same code to despread and recover the desired signal while treating other users' signals as wideband noise. Used in IS-95 (2G) and WCDMA (3G).
- **SDMA (Spatial Division Multiple Access):** Serves different users on the same frequency and time by using highly directional smart antennas (beamforming) to separate signals spatially.

TDMA Frame and Slot Structure



2. Modulation Schemes in Mobile Communications

Mobile radio channels require modulation schemes that are spectrally efficient, robust against multipath distortion, and power efficient (low peak-to-average power ratio).

2.1 BPSK, QPSK, and Variants

- **BPSK (Binary Phase Shift Keying):** Maps 1 bit per symbol to phases 0 and π . Simple and robust, but has low spectral efficiency (1 bps/Hz).
- **QPSK (Quadrature Phase Shift Keying):** Maps 2 bits per symbol to four phases (45° , 135° , 225° , 315°). Achieves double the spectral efficiency of BPSK (2 bps/Hz) within the same bandwidth.

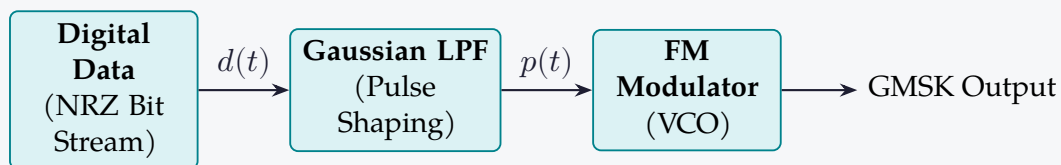
- **OQPSK (Offset QPSK):** Introduces a $T_s/2$ time delay between the in-phase (I) and quadrature (Q) bit streams. This prevents simultaneous transitions of both streams, limiting phase changes to a maximum of 90° (instead of 180° in standard QPSK). This reduces envelope fluctuations after filtering, allowing efficient non-linear amplifiers.
- **$\pi/4$ -DQPSK (Differential QPSK):** The phase constellation rotates by $\pi/4$ every symbol interval. The maximum phase change is limited to 135° , which prevents the signal envelope from passing through zero. This reduces spectral regrowth and simplifies demodulation using differential detection.

2.2 MSK and GMSK

Non-constant envelope signals experience severe distortion when passed through non-linear power amplifiers. Constant envelope modulation schemes solve this issue:

- **MSK (Minimum Shift Keying):** A continuous-phase frequency shift keying (CPFSK) scheme with a modulation index $m = 0.5$ (minimum frequency spacing for orthogonal detection). The phase changes linearly over each symbol period.
- **GMSK (Gaussian Minimum Shift Keying):** Derived from MSK by filtering the baseband NRZ data stream through a Gaussian low-pass filter before frequency modulation.
 - **Advantage:** The Gaussian filter smooths the phase transitions, suppressing the side lobes of the transmitted spectrum. This minimizes adjacent channel interference (ACI).
 - **Trade-off:** The smoothing introduces controlled Inter-Symbol Interference (ISI) in the time domain, requiring an equalizer at the receiver. Used in the GSM standard.

GMSK Transmitter Block Diagram



3. Multicarrier Modulation and OFDM

3.1 Basic Principles of OFDM

Orthogonal Frequency Division Multiplexing (OFDM) is a multicarrier modulation scheme that divides a high-speed data stream into multiple parallel, low-speed streams modulated onto orthogonal subcarriers.

Subcarrier Orthogonality: Two subcarriers are orthogonal if their frequencies are spaced at multiples of the inverse of the symbol duration ($1/T$). Mathematically:

$$\frac{1}{T} \int_0^T e^{j2\pi f_m t} \cdot e^{-j2\pi f_n t} dt = 0 \quad \left(\text{for } f_m - f_n = \frac{k}{T}, k \neq 0\right) \quad (1)$$

This allows their spectra to overlap in the frequency domain without causing mutual interference (ICI), maximizing spectral efficiency.

3.2 Mathematical Formulation (IFFT/FFT Relation)

An OFDM symbol is generated in the digital domain using the Inverse Fast Fourier Transform (IFFT). Let X_k be the complex modulation symbol (e.g., QAM) mapped to the k -th subcarrier. The discrete-time baseband OFDM signal is:

$$x(n) = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} X_k e^{j \frac{2\pi k n}{N}} \quad \Rightarrow \text{IDFT / IFFT operation} \quad (2)$$

At the receiver, the symbols are recovered using the Fast Fourier Transform (FFT):

$$X_k = \frac{1}{\sqrt{N}} \sum_{n=0}^{N-1} x(n) e^{-j \frac{2\pi k n}{N}} \quad \Rightarrow \text{DFT / FFT operation} \quad (3)$$

3.3 Cyclic Prefix (CP) Insertion

To eliminate Inter-Symbol Interference (ISI) caused by multipath delay spread, a **Cyclic Prefix (CP)** is appended to the beginning of each OFDM symbol.

- **Mechanism:** The CP is a copy of the end portion of the OFDM symbol, prepended to the start. The duration of the CP (T_{CP}) must be longer than the maximum delay spread of the channel ($\tau_{\max}$).
- **Purpose:**
 - Acts as a guard time, preventing the preceding symbol's multipath components from interfering with the current symbol (eliminates ISI).
 - Converts linear channel convolution into circular convolution, allowing simple one-tap frequency-domain equalization (FDE) at the receiver.

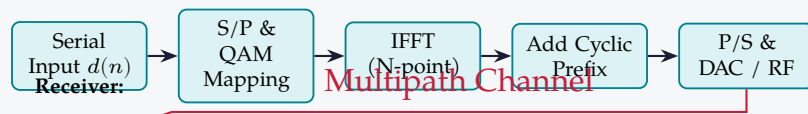
3.4 Peak-to-Average Power Ratio (PAPR) Issue

The primary drawback of OFDM is its high PAPR.

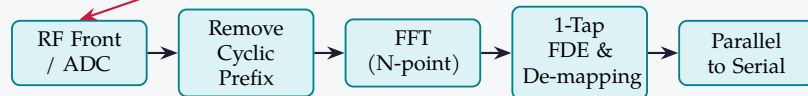
- **Cause:** Because an OFDM symbol is the sum of many independent subcarriers, their constructive phase alignment can produce large peak signals relative to the average power.
- **Impact:** High PAPR requires power amplifiers with wide linear ranges, reducing efficiency and increasing costs. Non-linear amplification causes out-of-band emissions and in-band distortion.
- **Mitigation:** Clipping and filtering, selective mapping (SLM), coding (e.g., Golay complementary sequences), and using Single-Carrier FDMA (SC-FDMA) on the uplink (as in LTE).

OFDM Transmitter and Receiver Block Diagrams

Transmitter:



Receiver:



Quick Revision — Module IV

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Key Formulas at a Glance

Subcarrier Frequency Spacing	$\Delta f = \frac{1}{T}$ (T = active OFDM symbol duration)
OFDM Subcarrier Orthogonality	$\frac{1}{T} \int_0^T e^{j2\pi f_m t} \cdot e^{-j2\pi f_n t} dt = 0$ ($f_m - f_n = \frac{k}{T}$)
Discrete OFDM Signal (IFFT)	$x(n) = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} X_k e^{j\frac{2\pi kn}{N}}$
Cyclic Prefix Condition	$T_{CP} \geq \tau_{\max}$ ($\tau_{\max}$ = max channel delay spread)
Peak-to-Average Power Ratio	$PAPR = \frac{\max x(t) ^2}{\mathbb{E}[x(t) ^2]}$
MSK Frequency Separation	$\Delta f_{\min} = \frac{1}{2T_b}$ (minimum for coherent orthogonality)

One-Line Definitions

- **FDMA:** Allocates unique, narrow frequency bands to users for the duration of the call.
- **TDMA:** Divides the carrier frequency into periodic time slots, assigning one to each user.
- **CDMA:** Allows users to share the entire bandwidth simultaneously using unique orthogonal codes.
- **Offset QPSK:** A QPSK variant that delays the Q-channel by half a symbol period to prevent envelope drops.
- **GMSK:** A continuous-phase modulation scheme that shapes baseband pulses using a Gaussian filter before modulation.
- **OFDM:** A multicarrier modulation scheme dividing a high-rate stream into orthogonal, overlapping subchannels.
- **Cyclic Prefix:** A copy of the end of an OFDM symbol prepended to the start to eliminate ISI.

Frequently Asked / Viva Questions

Q1. What is the primary difference between TDMA and FDMA regarding slot guard times?

TDMA requires guard times between time slots to prevent overlapping from propagation delays. FDMA requires guard bands in the frequency domain to prevent adjacent channel interference.

Q2. Explain the physical meaning of orthogonality in OFDM.

Subcarrier frequencies are spaced at multiples of the inverse of the symbol duration ($1/T$). At the peak of any single subcarrier's spectrum, the spectra of all

Frequently Asked / Viva Questions (contd.)

other subcarriers are zero, preventing inter-carrier interference.

Q3. Why does GMSK exhibit lower adjacent channel interference (ACI) than MSK?

GMSK uses a Gaussian low-pass filter to smooth the rectangular data pulses before modulation. This removes sharp phase transitions, suppressing the side lobes of the transmitted spectrum.

Q4. Explain the near-far problem in CDMA. How is it mitigated?

A mobile station near the base station can drown out weaker signals from distant users because all users share the same bandwidth. It is mitigated using rapid closed-loop power control.

Q5. What is the role of the Cyclic Prefix (CP) in OFDM? What happens if $T_{CP} < \tau_{max}$?

The CP acts as a guard time to absorb multipath reflections, eliminating ISI. If $T_{CP} < \tau_{max}$, multipath components from the preceding symbol will leak into the current symbol, causing ISI.

Q6. Why does OQPSK have less envelope fluctuation than conventional QPSK?

By offset-aligning the I and Q bit streams by $T_s/2$, both streams never change states simultaneously. This limits phase transitions to 90° , avoiding the 180° phase jumps that cause envelope drops in QPSK.

Q7. Define the Peak-to-Average Power Ratio (PAPR) in OFDM and list its mitigation methods.

PAPR is the ratio of peak signal power to average power. High PAPR occurs when subcarriers add constructively. Mitigation includes clipping/filtering, selective mapping (SLM), and SC-FDMA.

Q8. Describe GMSK's block diagram and identify where it is used.

It consists of digital NRZ data $\rightarrow$ Gaussian low-pass filter $\rightarrow$ FM modulator. It is used as the modulation standard for GSM (2G) systems.

Q9. What is the difference between orthogonal codes and pseudo-noise (PN) codes in CDMA?

Orthogonal codes (e.g., Walsh codes) have zero cross-correlation when perfectly synchronized, separating users on the downlink. PN codes have excellent autocorrelation, separating users on the uplink.

Q10. How does the IFFT generate an OFDM symbol?

The IFFT takes N complex frequency-domain symbols (QAM mapped) and converts them into N orthogonal discrete time-domain samples.

Q11. Explain why OFDM simplifies equalizer design at the receiver.

The cyclic prefix converts the linear channel convolution into circular convolution. This allows the receiver to equalize the channel using a single-tap multiplier per subcarrier in the frequency domain.

Frequently Asked / Viva Questions (contd.)

Q12. What is the modulation index of MSK and why is it special?

The modulation index is $m = 0.5$. This represents the minimum frequency separation ($\Delta f = 1/2T_b$) that allows two FSK signals to be detected coherently and remain orthogonal.

Comparison of Multiple Access Schemes

Feature	FDMA	TDMA	CDMA
Separation	Frequency band.	Time slot.	Orthogonal code.
Bandwidth	Divided into narrow channels.	Shared in groups.	Shared across entire band.
Handoff Type	Hard handoff.	Hard handoff.	Soft handoff.
Limitations	Guard bands, filter cost.	Guard times, synchronization.	Near-Far problem, power control.